MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING, DIRECTIONAL

16 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 149.4 and 149.5

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

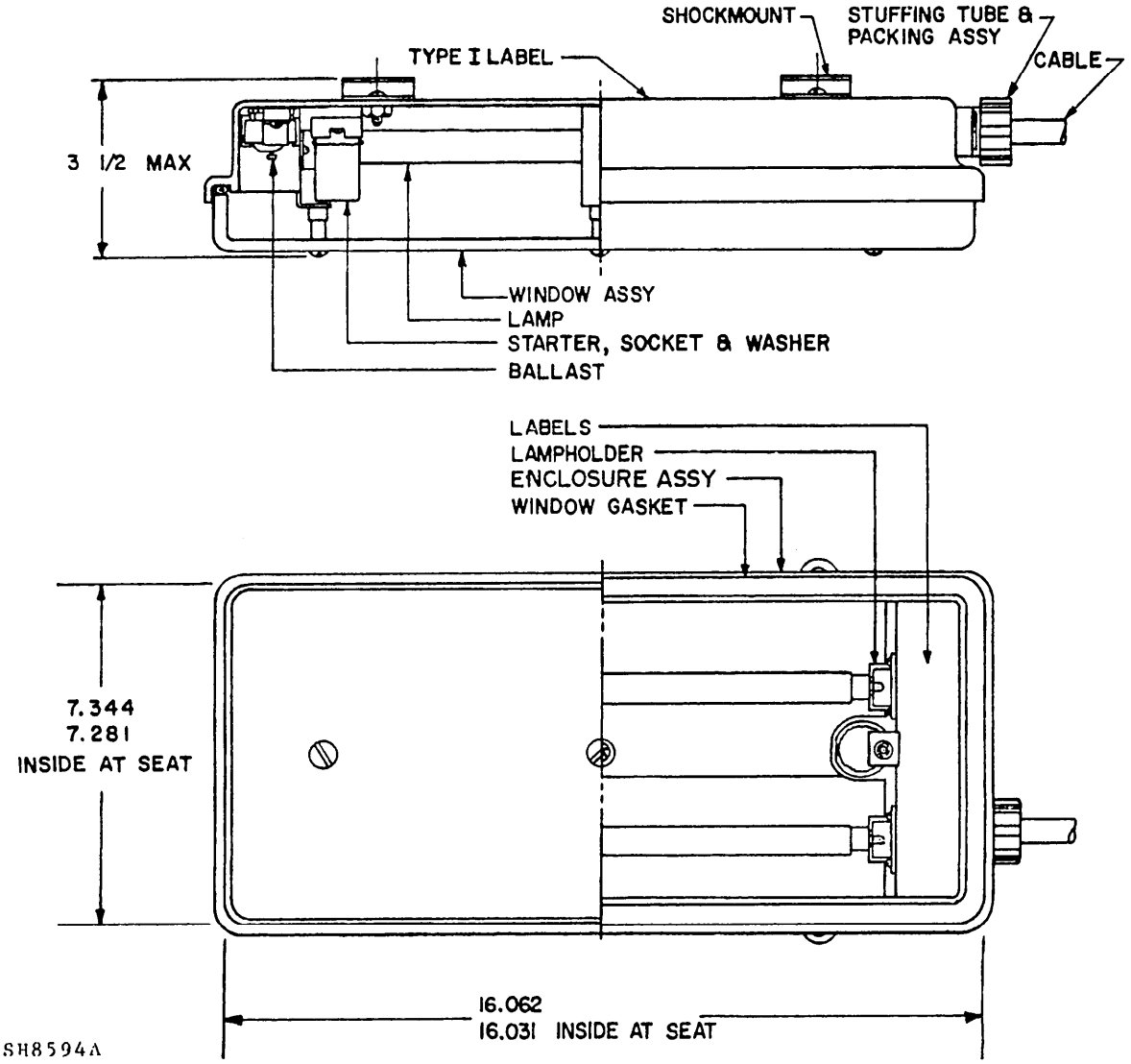


FIGURE 1. Typical view of Symbol 149.4 and 149.5.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

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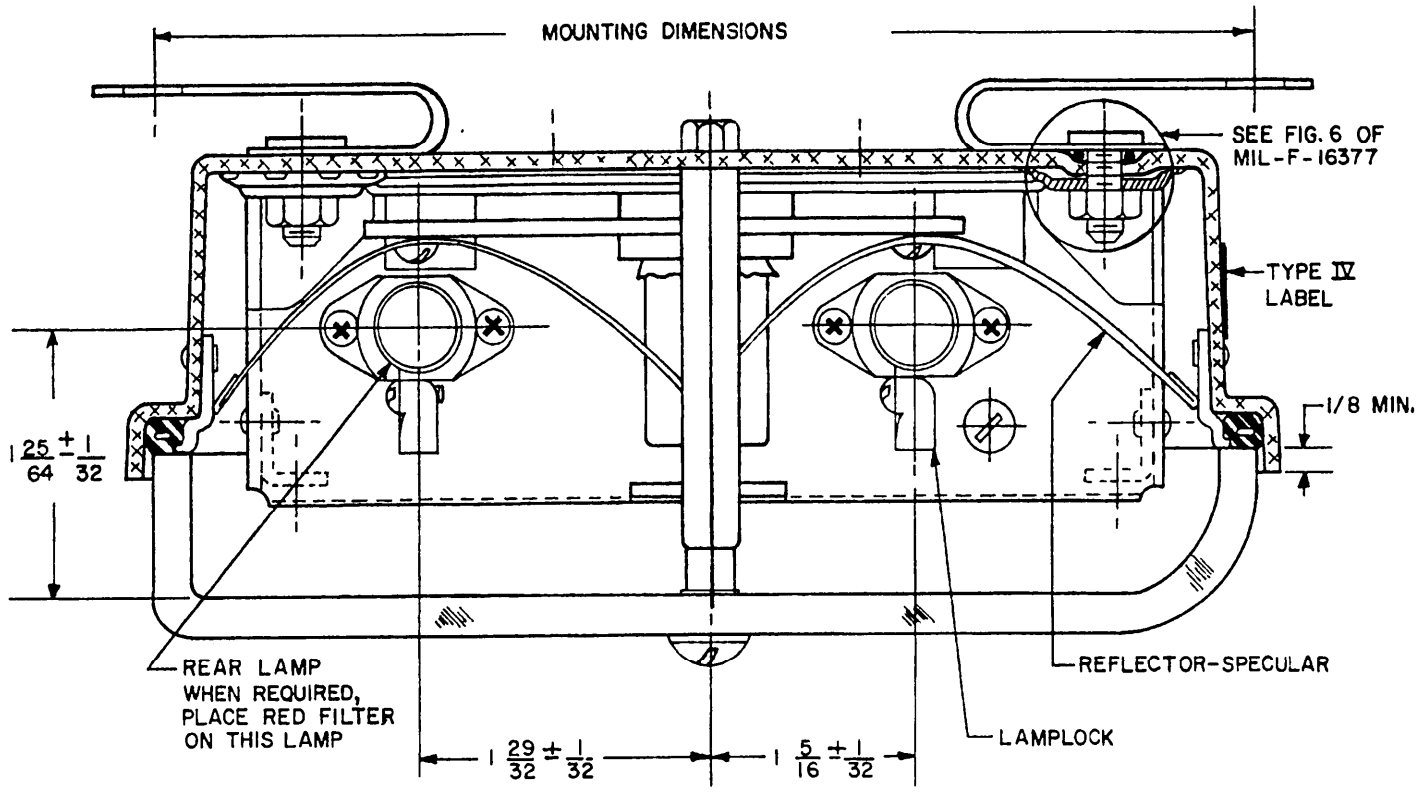


FIGURE 2. Typical section for Symbol 149.4.

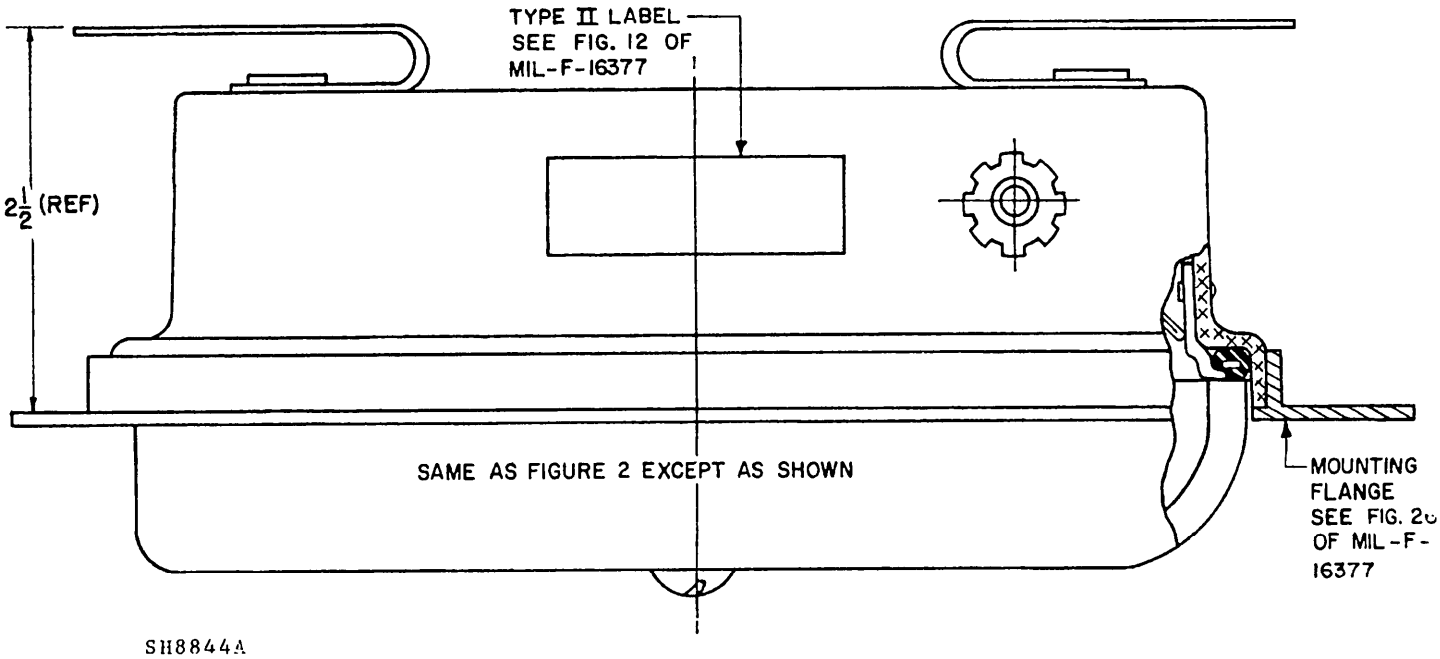
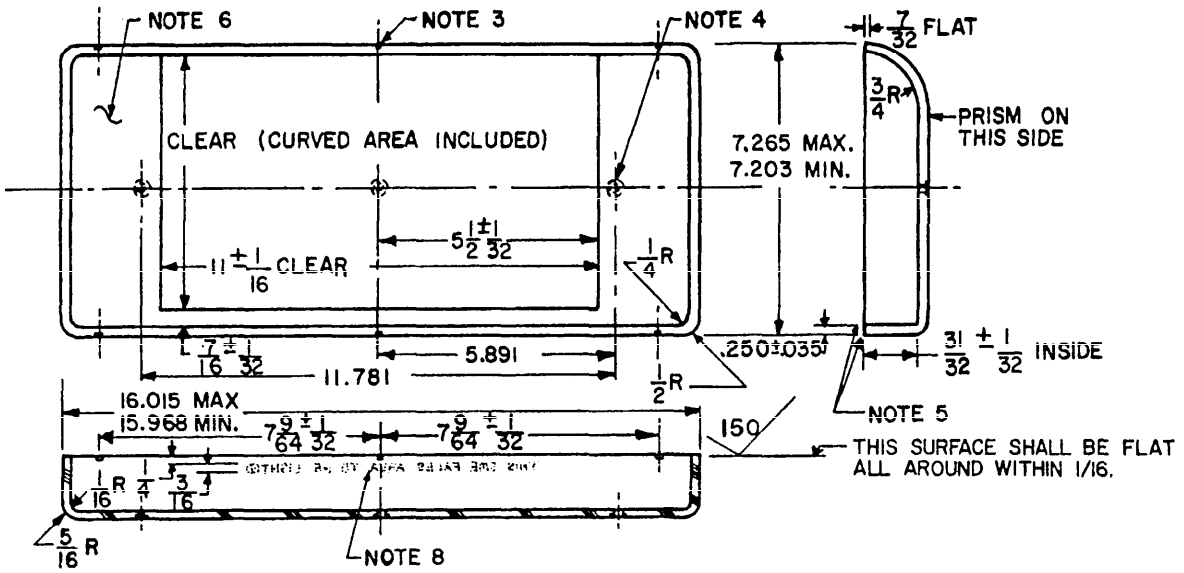
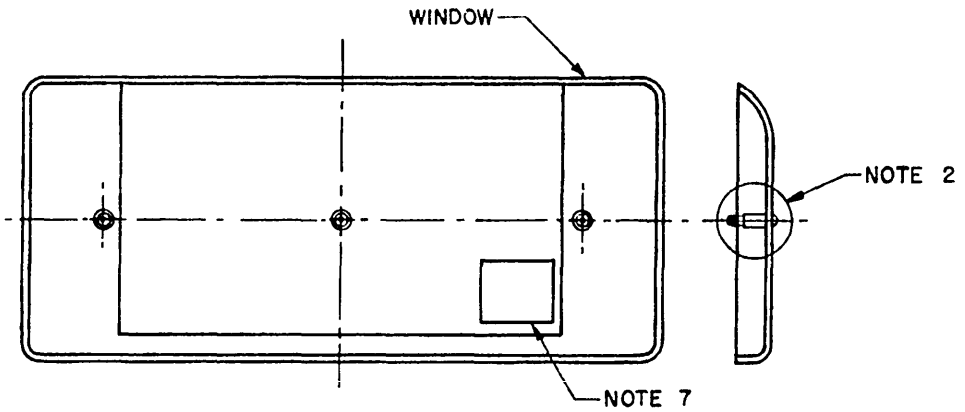


FIGURE 3. Typical section for Symbol 149.5.

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WINDOW



WINDOW ASSEMBLY

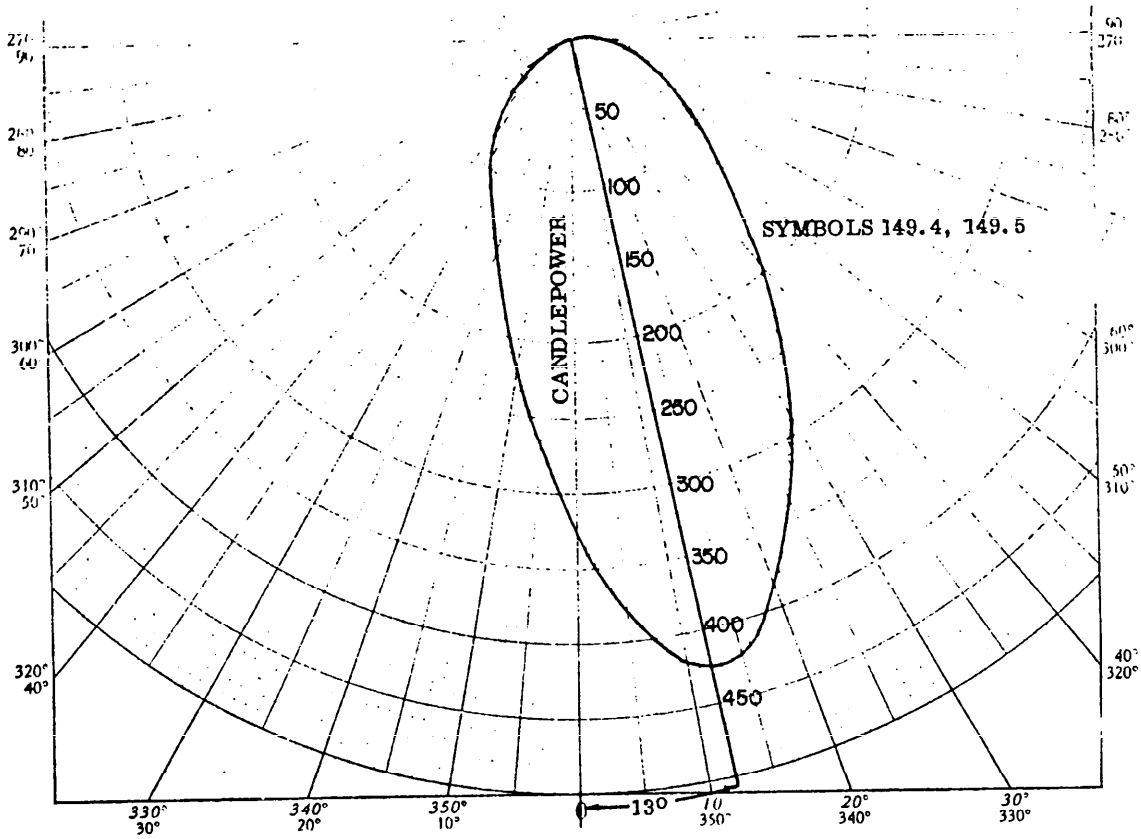
SH8846A

NOTES:

1. Material for the window shall be 1/4 inch thick, clear prismatic acrylic plastic, Type CLEAR-2 of MIL-P-24191. The prisms shall be on the outside surface of the windows.
2. Three short window securing screw assemblies shall be installed in accordance with 3.7.4.3 of MIL-F-16377.
3. When required, six notches (3 each side) shall be filed as shown in accordance with 3.7.4.4 of MIL-F-16377.
4. Drill 0.191 diameter hole and counterbore opposite side 0.323 diameter, 0.050 deep. Break sharp edges on the inside surface of all holes.
5. Break sharp edges 1/64 of an inch around entire window.
6. The inside surface of the window except the area indicated as CLEAR shall be finished white in accordance with 3.8.4.1 of MIL-F-16377.
7. When required, one Type VI label shall be applied on the inside surface of the window.
8. "THIS SIDE FACES AREA TO BE LIGHTED" shall be stenciled on the inside surface of the curved area readable from outside surface.

FIGURE 4. Window assembly, clear and white prismatic - directional.

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*ANGLE	CANDLEPOWER
	SYM 149.4, 149.5
0	420
5	400
10	355
15	300
20	235
25	185
30	155
35	132
40	110
45	95
50	80
55	60
60	55

*ANGLE	CANDLEPOWER
	SYM 149.4, 149.5
65	40
70	30
75	25
80	15
85	5
90	0
95	0
100	0
105	0
110	0
115	0
120	0
125	0

* Measured at an angle of 13° from vertical.

Photometric test data:

Distance: 10 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 380 lumens per lamp.

FIGURE 5. Candlepower distribution curves.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:

149.4: Complete lighting fixture assembly, standard mounting with two 8-watt cool white fluorescent lamps and clear and white prismatic directional window assembly.
149.5: Complete lighting fixture assembly, flush mounting, with two 8-watt cool white fluorescent lamps and clear and white prismatic directional window assembly.

DESIGN:
Electrical rating: 16 watts, 120 volts, 60 Hertz.
Dimension and configurations: See figures 1, 2, and 3 and figure 10 of MIL-F-16377.
Photometric data: see figure 5.
Lampholder spacing: see figure 21 of MIL-F-16377.
Wiring diagram: see figure 23 of MIL-F-16377.
Mounting dimensions:
Symbol 149.4 : see figure 25 of MIL-F-16377.
Symbol 149.5 : see figure 26 of MIL-F-16377.
Window assembly: : see figure 4.
Enclosure: Watertight. See 3.5.11 of MIL-F-16377.
Cable: 10 feet, type SJO or SJT construction in accordance with J-C-580 except AWG size 16, three conductors (white, black and red).
Weight:
Symbol 149.4 : 7 lbs. 8 oz. maximum.
Symbol 149.5 : 8 lbs. 12 oz. maximum.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figures 1 and 3:

(a) Two type I
(b) One type II for Symbol 149.5 only.
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY". List all standard parts shown in table I and window assemblies of table IV by name and stock number only.
(d) One type IV applied as shown in figure 2, bearing legend "THIS SIDE FACES AREA TO BE LIGHTED."

Standard Parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE:

First article inspection:
Complete fixtures: See table II
Window assemblies: See table III
Quality conformance:
Comparison inspection: Same as the first article inspection
Inspection of product for delivery:
Examination: See MIL-F-16377
Examination for preparation of delivery: MIL-F-16377
Quality conformance tests: See MIL-F-16377

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

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TABLE I. Standard parts.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/3	6240-00-299-5546	Industry Number F8T5/CW
Lamp lock	M16377/40-001	6250-00-399-7039	
Lampholder	MIL-L-970/10	6250-00-377-8590	
Startersocket	MIL-L-970/13	6250-00-295-2738	
Starter	8W, type III of W-S-755	6250-00-299-5962	Commercial Designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	
Shockmount	M16377/41-009 or M16377/41-010	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-F-16377
Red filter tube	M16377/42-001	6210-00-399-6297	Not furnished See 3.7.7 of MIL-F-16377

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

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TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

TABLE IV. Part number and general information.

Military part 1/ number M16377/6-	Item name	National Stock Number (NSN)	Remarks
149.4	Complete fixture, Symbol 149.4	6210-00-548-0226	New
149.5	Complete fixture, Symbol 149.5		New
002	Window assembly, clear and white prismatic, directional		New

1/ Military part numbers M16377/6-149.2, 149.3 and 001 have been superseded by Military part numbers M16377/6-149.4, 149.5 and M16377/67-005 respectively. Superseded fixtures were previously covered by revision -- of this specification, MS17181 and NAVSHIPS Drawing 9000-S6401-74459.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
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